Design Netflix

SystemsExpert design walkthrough

Clarifying questions

- Focus on streaming/video delivery, recommendations, or both?
- How many titles and subscribers?
- Need to support many device types and bitrates?

Functional requirements

- Browse catalogue, personalised home page
- Stream a title with adaptive bitrate
- Resume playback across devices

Non-functional requirements

- Very high availability, global low latency
- Startup < 2 s, minimal rebuffering

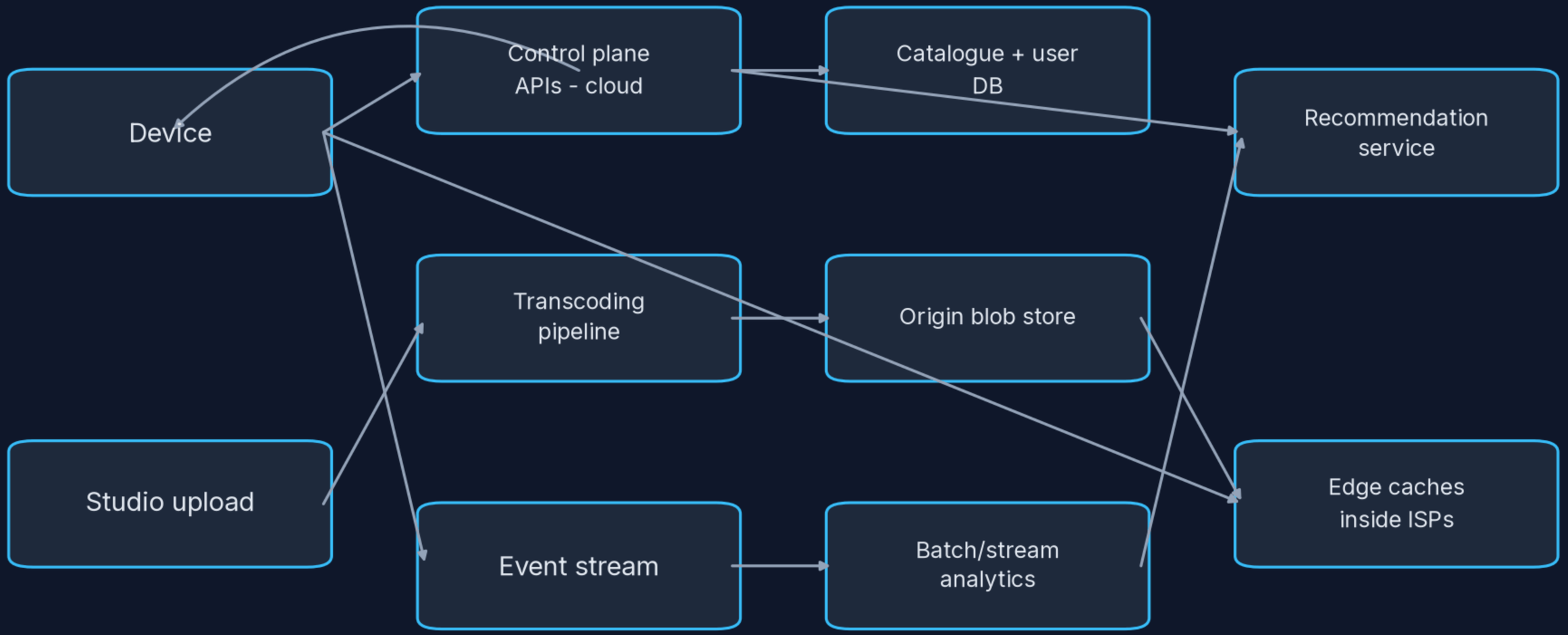
Capacity estimates

- ~200M subscribers, ~10k titles; each title encoded in ~dozens of bitrate/resolution variants → PB scale
- Peak egress tens of Tbps → must be served from edge caches (Open Connect style) close to ISPs

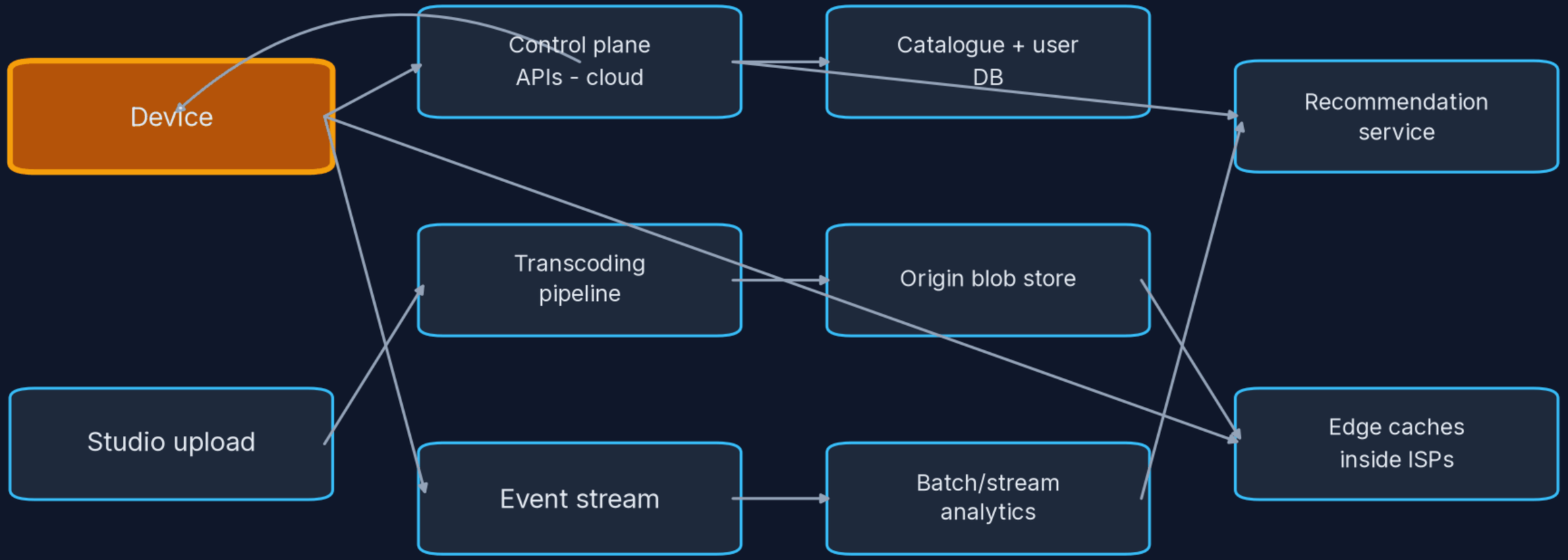
Capacity: worked out

- Peak 10M concurrent streams × 5 Mbps
- = 50 Tbps of egress
- Edge box ~100 Gbps → ~500+ boxes at peak
- Catalogue 10k titles × 50 encodes × ~5 GB ≈ PBs

High-level architecture

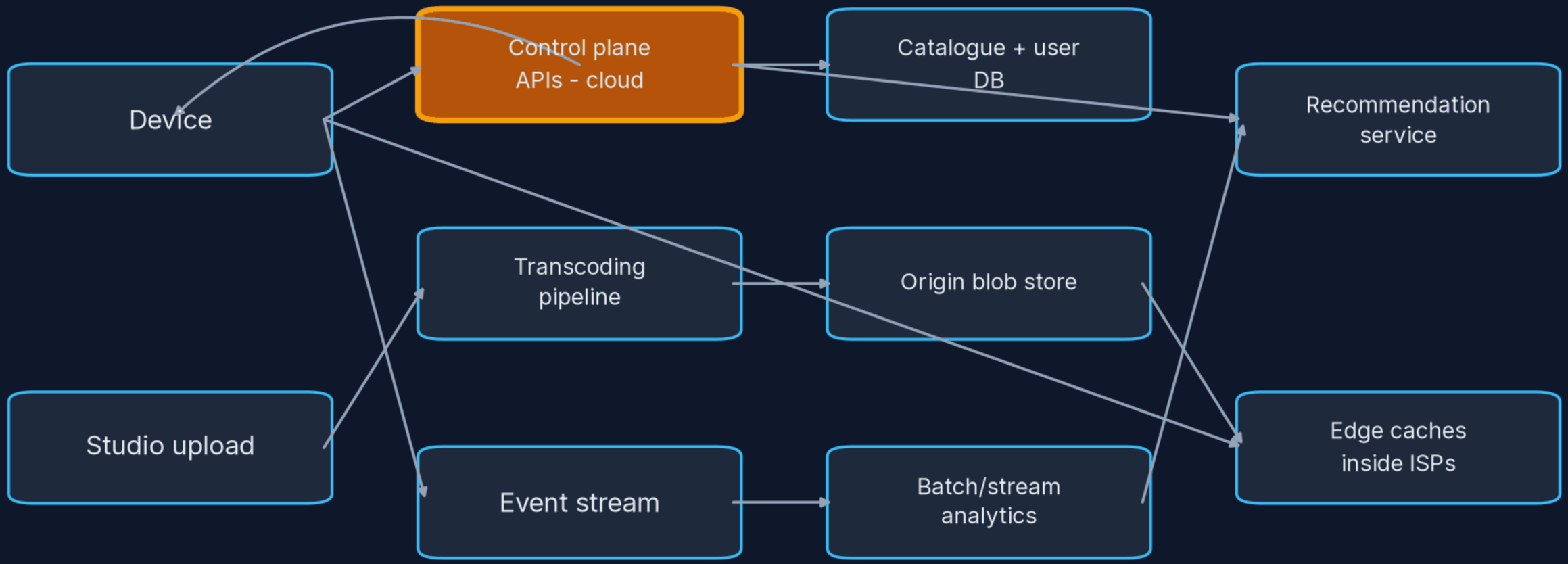


Request flow: Press Play



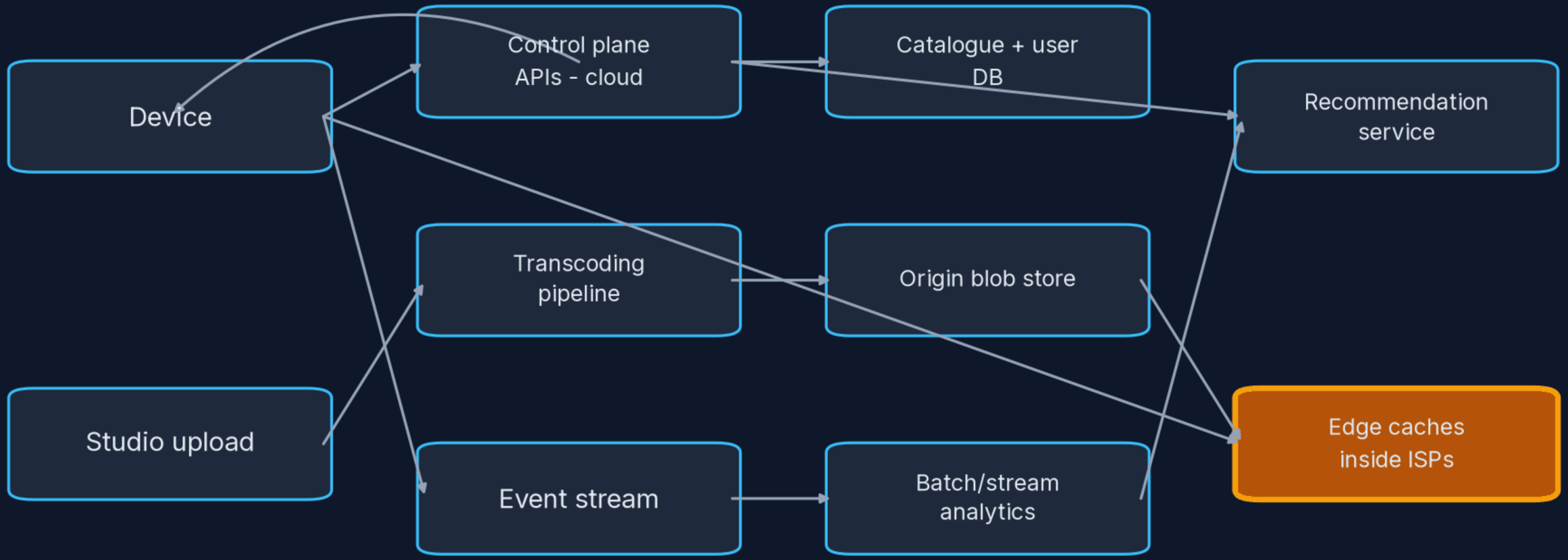
1. User presses Play

Request flow: Press Play



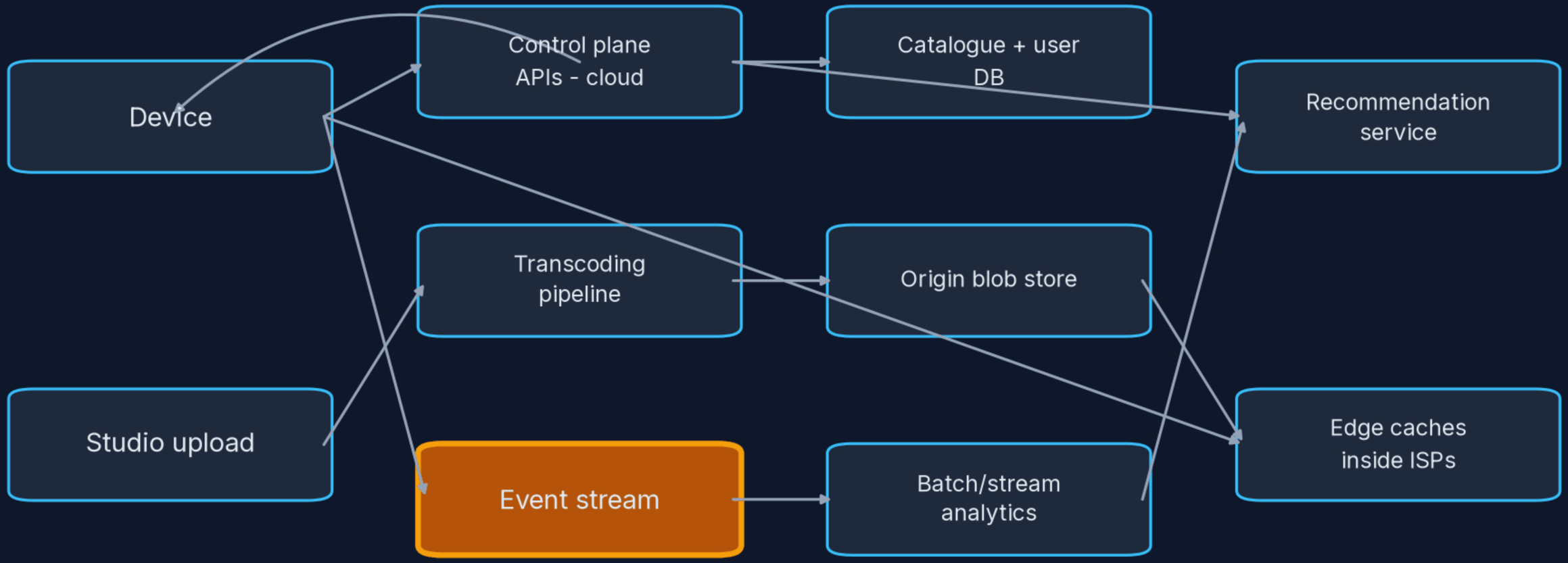
2. Control plane picks a CDN

Request flow: Press Play



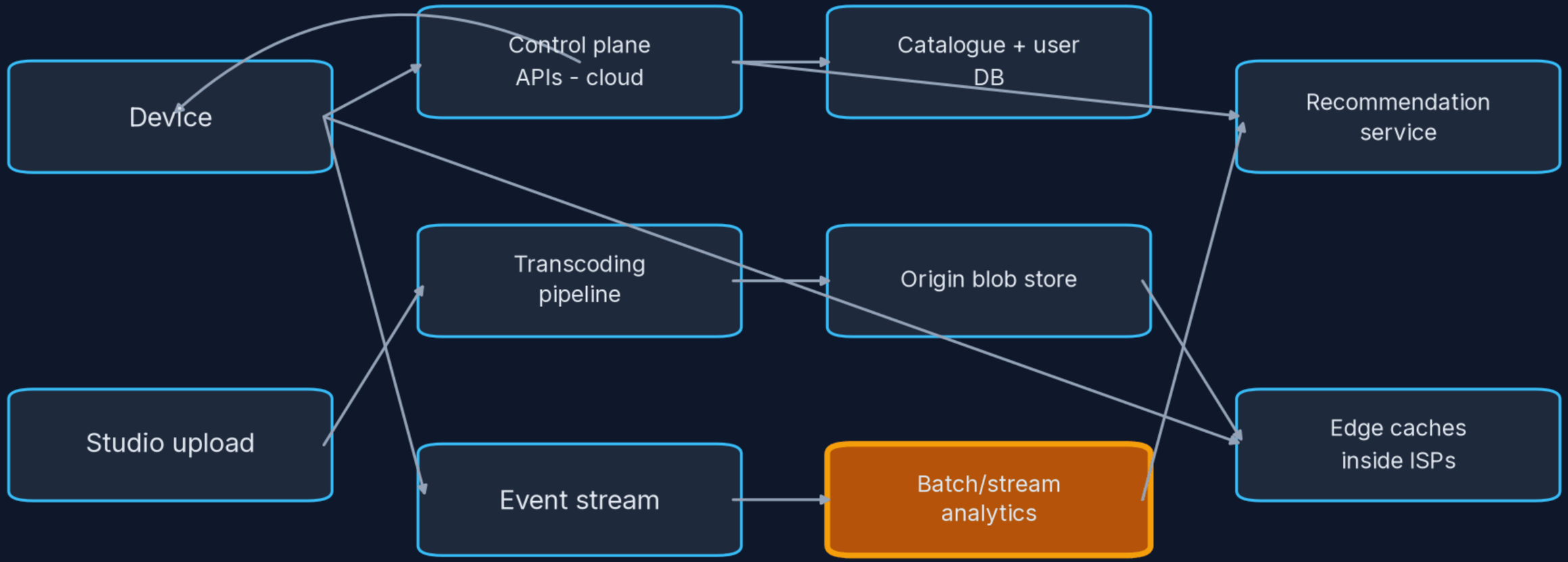
3. Stream segments from the edge

Request flow: Press Play



4. Heartbeats record progress

Request flow: Press Play



5. Analytics feed recommendations

Data model

- Titles(id, metadata, encodes[])
- Viewing_history(user_id, title_id, position, updated_at) - write-heavy, wide-column store
- Manifest: per title, list of segment URLs per bitrate

API

```
GET  /home                          -> personalised rows
GET  /titles/{id}
POST /playback/{titleId}/start   -> manifest + edge URLs + DRM license info
POST /playback/{titleId}/heartbeat {position}
```

Encoding

- Split source into chunks, transcode in parallel into many codecs/bitrates; segment into ~2-4 s pieces for ABR (DASH/HLS).

Delivery

- Popular content is pre-positioned at edges during off-peak; clients measure throughput and switch bitrate per segment.

Recommendations

- Offline batch training on viewing logs produces candidate lists; online service re-ranks per request.

Key decisions

- Own CDN inside ISPs → cost and quality at this egress
- Pre-position popular titles off-peak → cache hits at prime time
- Adaptive bitrate segments → smooth playback on any network
- Control plane in cloud, data plane at edge → each scales independently

Trade-offs & alternatives

- Own CDN gives control and cost savings but large capex
- Pre-positioning wastes some storage on titles that won't be watched
- Resume position is eventually consistent (heartbeat every ~10-30 s)

Recap

- Clarify scope, then functional + non-functional needs
- Estimate load to size storage and fan-out
- Core pieces: Device, Control plane APIs - cloud, Catalogue + user DB, Recommendation service, Edge caches inside ISPs
- Know your trade-offs and failure modes